Customer Shopping Behavior Analysis

Uncovering insights for business growth.

Made with GAMMA

Project Overview

Goal

Analyze 3,900 customer purchases.

Focus

Spending patterns, segments, age groups, subscriptions.

Outcome

Guide business decisions and growth.

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Dataset Summary

Data Snapshot

- Rows: 3,900
- Columns: 18
- Missing Data: 37 values in 'Review Rating'

Key Features

- Customer Demographics (Age, Gender, Location, Subscription)
- Purchase Details (Item, Category, Amount, Size, Color, Season)
- Shopping Behavior (Shipping, Discount, Payment, Frequency, Rating)

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Data Processing in Python

01

Data Loading & Inspection

Pandas for import, df.info() & .describe() for structure.

02

Missing Data Handling

Filled 37 null 'Review Rating' values with median per category.

03

Column Standardization

Renamed columns to lowercase for readability.

04

Feature Engineering

Created 'age_group' and 'purchase_frequency_days'.

05

Data Consistency

Dropped redundant 'Promo Code Used' column.

06

Database Integration

Loaded cleaned data into PostgreSQL for SQL analysis.

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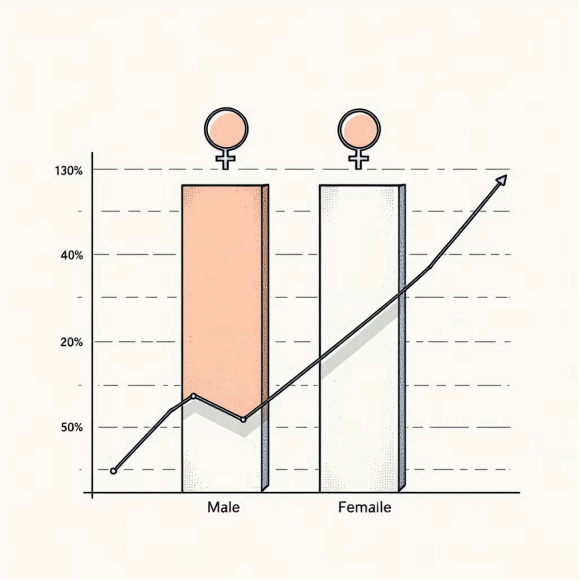
SQL Analysis: Revenue & Discounts

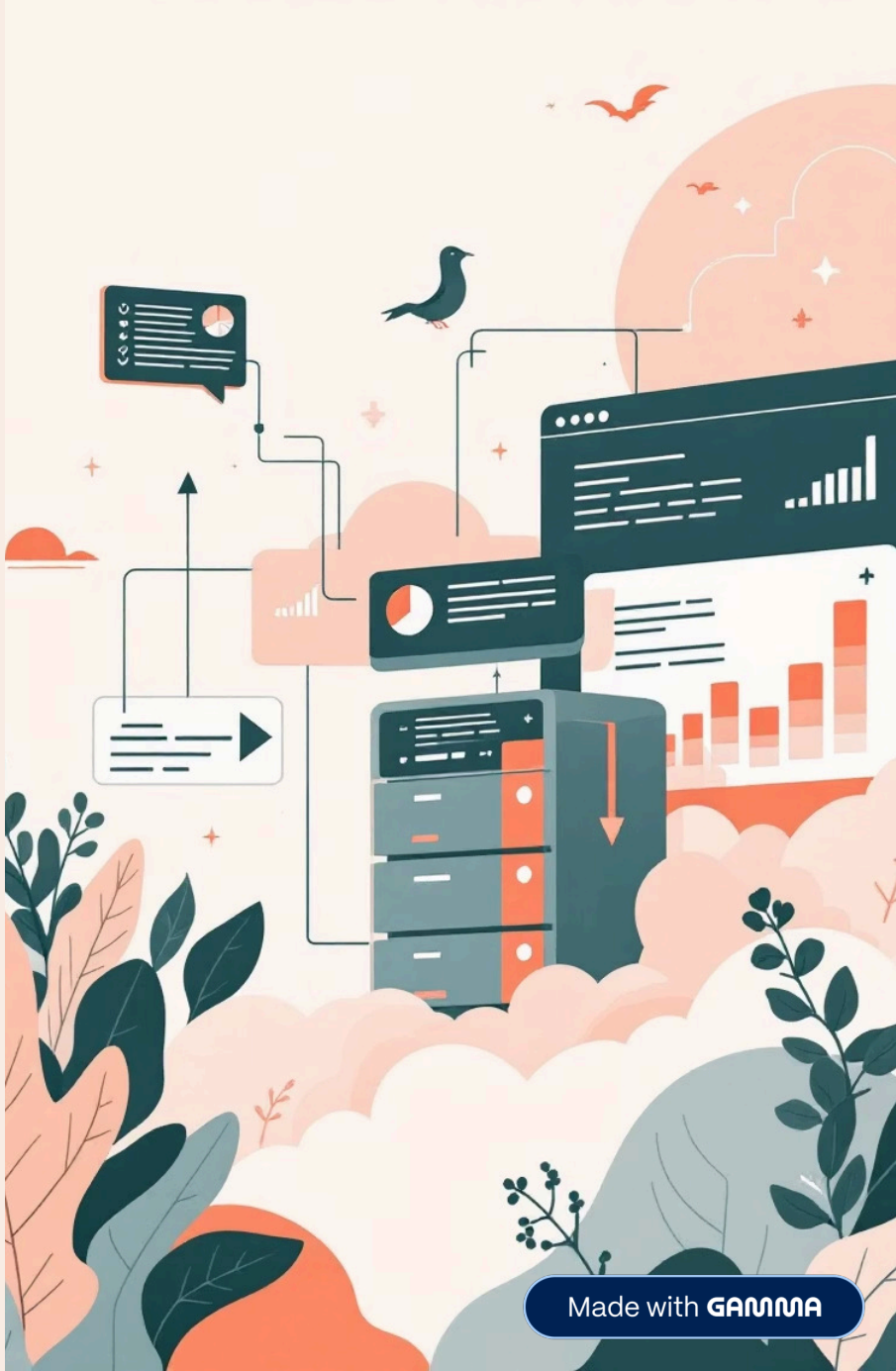
Gender Revenue

Total revenue by male vs. female customers.



Discount Spenders

Customers using discounts but spending above average.



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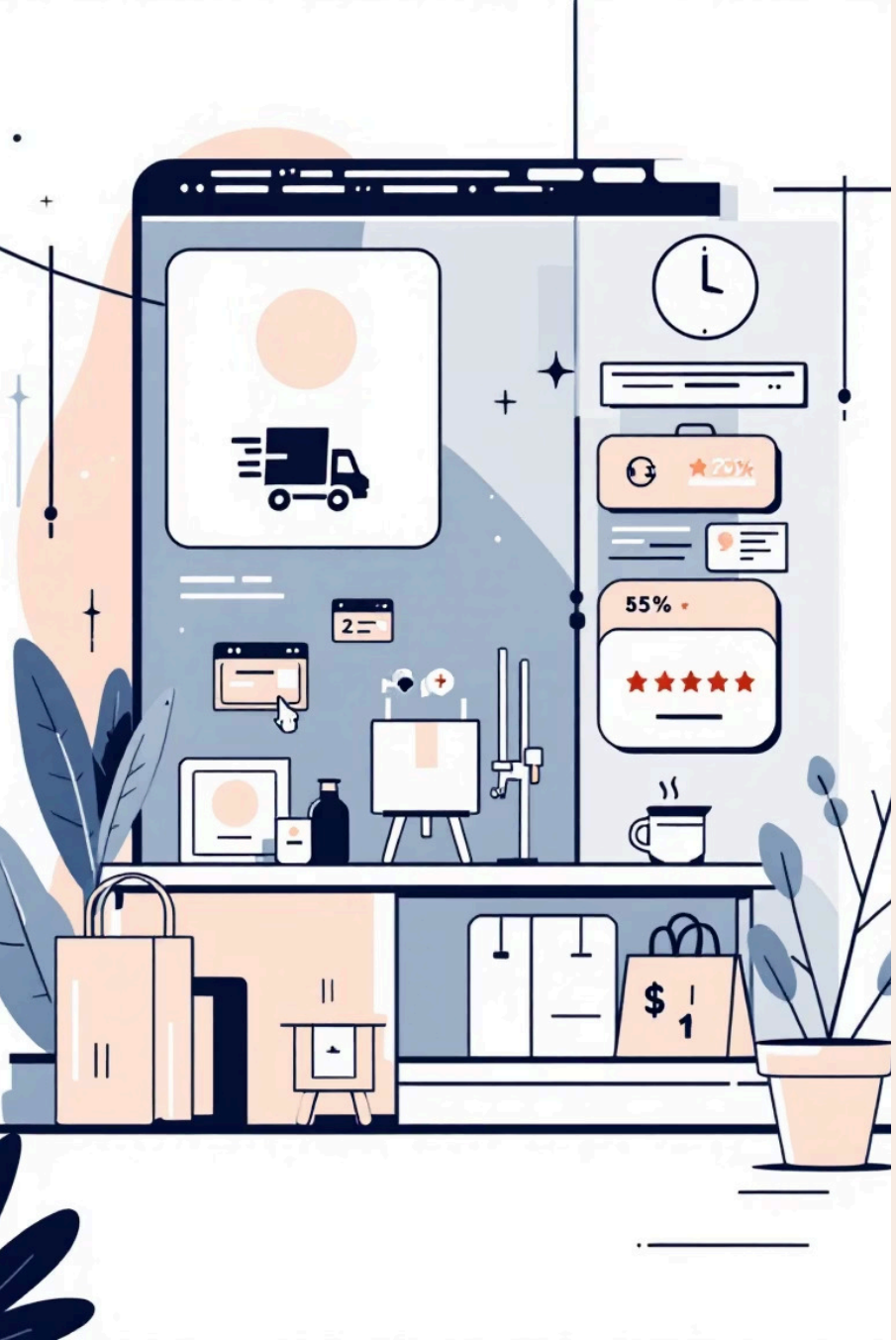

SQL Analysis: Product Performance

1

Top 5 Products

Highest average review ratings.

2

Shipping Impact

Average purchase amount: Standard vs. Express.

3

Discounted Products

Top 5 products with highest discount purchase percentage.

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SQL Analysis: Customer Segmentation

New Customers

First-time buyers.

Returning Customers

Multiple purchases.

Loyal Customers

Frequent, high-value buyers.

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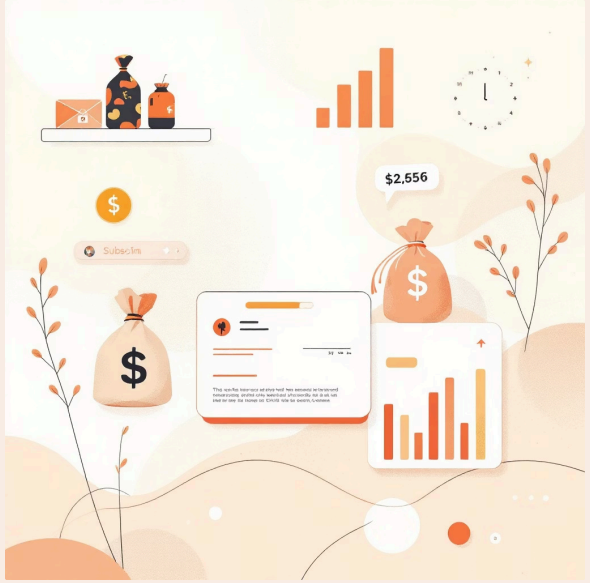
SQL Analysis: Subscription & Age

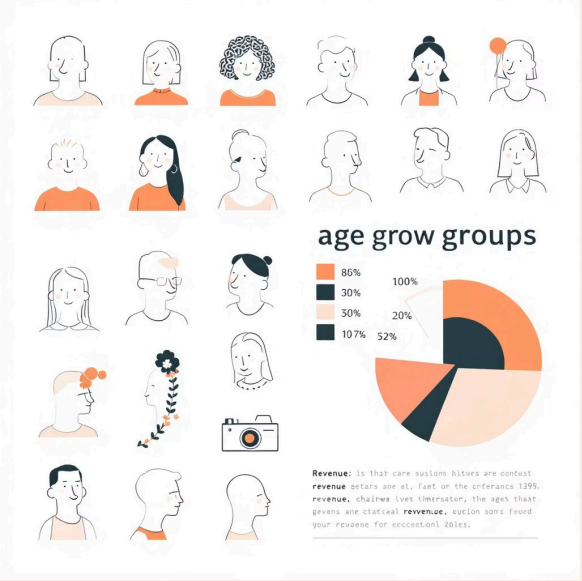
Subscriber Spending

Compare average spend and total revenue: subscribers vs. non-subscribers.



Age Group Revenue

Revenue contribution by each age group.



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SQL Analysis: Product & Loyalty

Top 3 Products

Most purchased within each category.

Repeat Buyers & Subscriptions

Are customers with >5 previous purchases also subscribers?

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Power BI Dashboard

Interactive visualizations for key insights.

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